

Worksheet 15

Variable frequency drive strategies

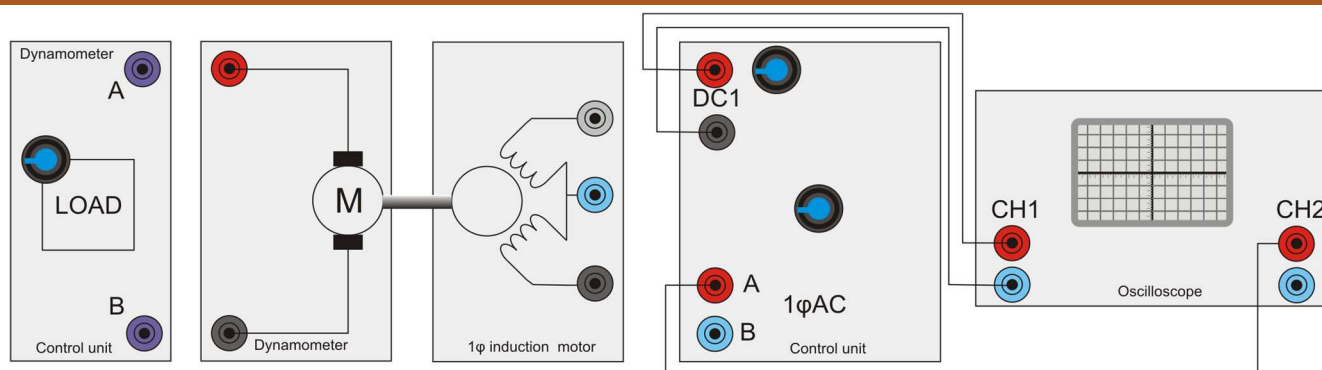
Modern electrical machines



In factories, induction motors are often used at fixed speeds, dictated by the 50 or 60Hz mains supply.

Where speed control is needed it requires a change in strategy.

This photograph shows variable frequency drives used in a mineral water plant.



Do not use a PC-based oscilloscope.
Use a traditional 'scope with no USB connection to the PC.

Over to you:

- 1) Set up the single-phase induction motor as shown in the diagram above.
 In this exercise, we are just looking at signals and so the motor and dynamometer do not need to be connected. However, the motor must be in the dynamometer cradle in order to activate the power supply .
- 2) On your PC run 'Open Control_1Phase.bat'.
- 3) Connect the oscilloscope as shown. In this application, the DC1 output gives a pulse at the start of each cycle. This can be used to trigger the scope.
- 4) On the next page, sketch the waveform of the single-phase AC output A.

So what?

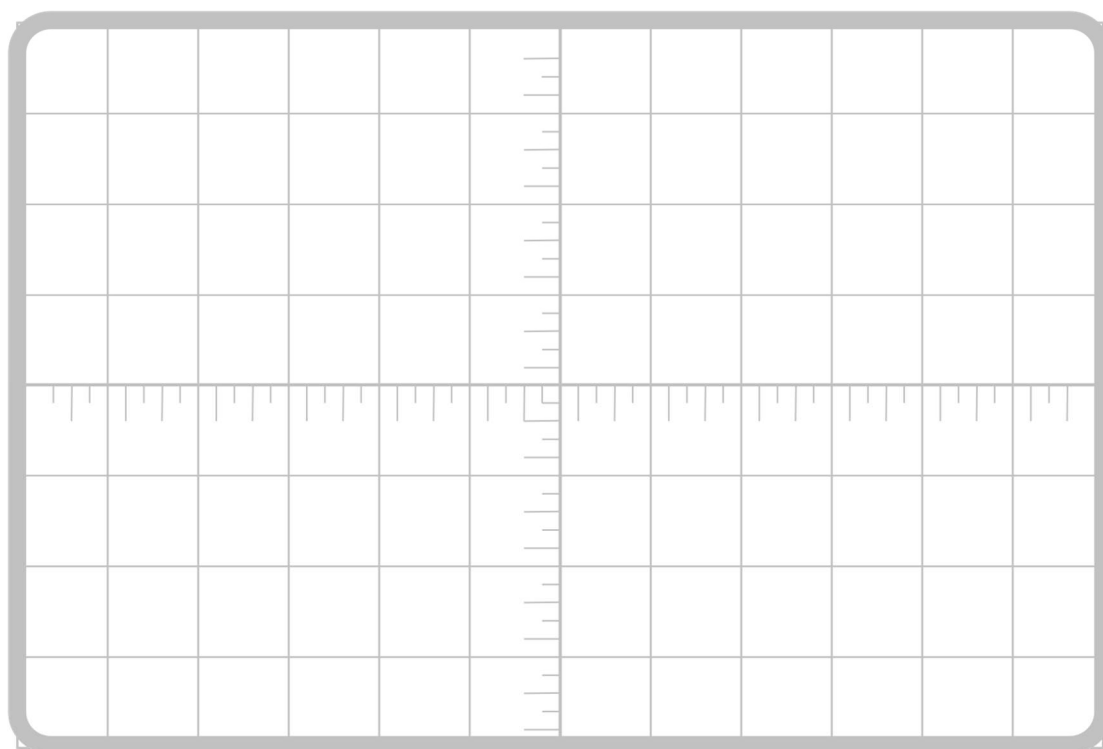
This exercise illustrates how PWM (Pulse Width Modulation) is used to create a waveform with variable pulse width but constant period, the period of the fundamental waveform.

The motor drive coils act as a low-pass filter for AC signals, smoothing out the waveform. It ends up looking much more like a traditional sine wave.

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Over to you:

With the dynamometer load at around 50%, measure and record in the table below the speed, torque, output power and input power, for frequencies of 50, 60 and 70Hz.

AC drive frequency	Speed in rpm	Torque	Output power	Input power
50Hz				
60Hz				
70Hz				

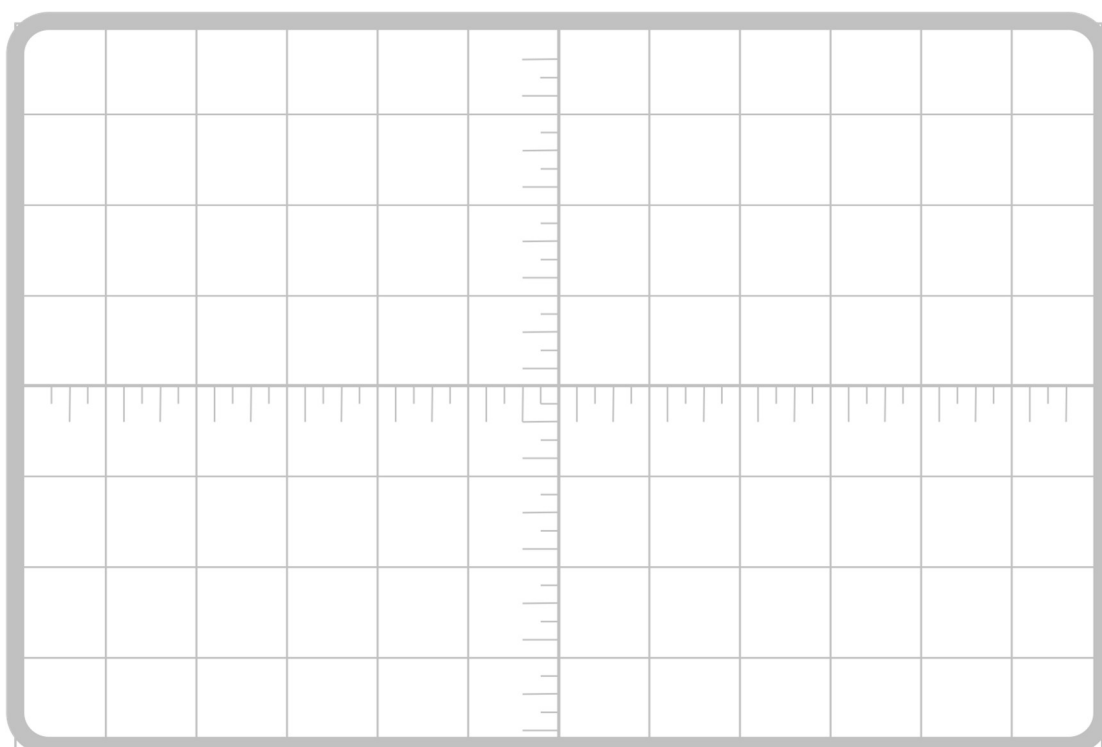
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Over to you:

- 1) In the software application, click on the button marked 'Sine' . This changes the output of the drive from pseudo-sine wave to a simple digital one.
- 2) Sketch the output at **A** when using the trigger pulse on DC1. (You could compare the signal from **B**.)
- 3) Then, with the dynamometer load at around 50%, measure and record in the table below the speed, torque, output power and input power, for frequencies of 50, 60 and 70Hz.



AC drive frequency	Speed in rpm	Torque	Output power	Input power
50Hz				
60Hz				
70Hz				

So what?

It is surprising how well a simple digital drive signal works for the induction motor. However, when energy conservation is paramount, the cost of a few bits of silicon needed to create the drive circuit is of secondary importance.

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Over to you:

- 1) In the block diagram of the control unit in the 'Reference' section, the motor drive outputs are shown as 'tri state'. This means that each output voltage can be high, low or open circuit.

The single phase 24V maximum outputs **A** and **B** are alternately positive and negative, effectively creating a maximum peak-to-peak voltage of 48V across the motor.

- 2) Obtain the waveforms and sketch them on the axes below.

You can use the on board oscilloscope here to look at the current waveform.
However the on-board scope is not fast enough to show PWM accurately.

